

PLAYGROUND DESIGN PROPOSAL

ASSOSA PRIMARY AND MIDDLE SCHOOL

SITE PARAMETERS

School	4 of 6
Total Students	230
Compound Area	50,662 m ²
Play Zone	≈ 550 m ²
Buildings	22 (4,048 m ²)
Open Space	46,614 m ²

EXISTING EQUIPMENT

- 2 Slides – Functional (need paint)
- 1 Balance Beam – Functional
- 2 Swings – 1 functional, 1 maint.
- 2 Ladders – Functional

EQUIPMENT GAPS

- 1 Merry-Go-Round – Nonfunctional (Remove)
- See-Saw – Missing
- Tunnel Crawl – Missing
- Stepping Stones – Missing

Site analysis, concept design, and detailed design proposal for pre-primary playground infrastructure serving 230 children aged 5 to 6 years within a 50,662 m² school compound. The pre-primary play zone of approximately 550 m² is a dedicated section within the largest primary and middle school campus, designed to provide a safe, stimulating play environment separate from older students.

1.1 Site Context Report

Assosa Primary and Middle School is the largest school in the IRC SUIDAC programme, located in the capital of Benishangul-Gumuz Regional State, western Ethiopia, at an elevation of approximately 1,570 metres above sea level. The school compound occupies a substantially fenced parcel of **50,662 m²** — the largest of all six programme schools — within a densely built-up urban neighbourhood in Assosa city. The compound is bounded by a 44-vertex fence polygon enclosing twenty-two permanent structures, with road access via the main entrance along the eastern boundary.

The site currently serves **230 pre-primary students** (124 male, 106 female) aged approximately 5 to 6 years, making it the highest-enrolment school in the programme. Twenty-two permanent buildings with a combined footprint of **4,048 m²** form a full primary and middle school campus. Buildings are primarily single-storey rectangular blocks with corrugated metal roofs, clustered in 2–3 rows along the western and northwestern boundaries. Despite the extensive building programme, the site retains **46,614 m² of available open space** — an exceptionally generous ratio of open space to built footprint.

The existing play equipment inventory includes two functional slides (requiring repainting), one functional balance beam, two chain swings (one functional, one requiring maintenance), and two functional ladders. A single merry-go-round has been assessed as **nonfunctional and unsafe** due to structural deterioration; it will be removed prior to new construction. Three categories of equipment are entirely absent from the site — see-saws, tunnel crawls, and stepping stones — representing critical gaps in the play value portfolio that must be addressed in the new design.

A notable feature of the site is the **exceptional tree canopy coverage of 60–70 percent**, as identified through VLM satellite image interpretation. This dense canopy of mature trees provides extensive natural shade, significantly reducing surface temperatures across the compound. The tree canopy is partially concentrated along the western and northwestern boundaries adjacent to building clusters, with additional scattered coverage across the open spaces. This exceptional canopy asset has been flagged for preservation and integration into the playground design as a natural shade infrastructure.

No on-site photographs were provided for this school, and a site survey visit is flagged as a priority requirement. The KML site plan data includes 22 building footprints, 28 line features (representing internal pathways, fence segments, and drainage features), and 16 point features (representing trees, utility access points, and existing equipment locations). All spatial planning in this document is based on the KML boundary polygon and VLM image interpretation, pending confirmation through field survey.

The site is characterised by lateritic red soils typical of the Ethiopian highland plateau. Climate data for Assosa indicates a mean annual temperature of 22 °C, with daily maxima occasionally exceeding 35 °C during the dry season. Annual rainfall averages 1,200 mm, concentrated in the kiremt (main rainy) season from June to September. The combination of extensive tree canopy, large open space, and gentle topography makes this site ideally suited for a comprehensive playground development.

1.2 Constraint Map Description

The site has been divided into four colour-coded constraint zones based on the overlay analysis of the KML boundary data, VLM image interpretation, and 16 KML point features. These zones guide all subsequent layout decisions and equipment placement within the largest compound in the programme.

Zone	Colour	Description	Area (m ²)	Constraint
RED — No-Go	Red	Fence setback zone (1.5 m from 44-vertex fence polygon), building wall setbacks (3 m for swings/slides), road buffer along eastern entrance. Includes nonfunctional merry-go-round removal zone.	≈ 2,200	Hard exclusion
YELLOW — Caution	Amber	Existing equipment zones requiring retrofit (2 slides, 2 swings, 1 balance beam, 2 ladders), dense tree canopy root protection zones (1 m radius from trunks across 60–70% canopy area). Equipment modifications require engineer sign-off.	≈ 4,800	Conditional use
GREEN — Available	Green	Open central and eastern areas cleared of tree canopy, cleared spaces between building clusters, and dedicated expansion zones. Primary targets for the ~550 m ² pre-primary play zone.	≈ 36,000	Preferred
BLUE — Infrastructure	Blue	Internal pathway corridors (28 line features from KML, 1.2 m min width), drainage swales, utility points (16 KML points), and main entrance road. Must remain clear for access and water flow.	≈ 3,600	Service reserve

Table 1. Constraint zone classification and area allocation.

1.3 Opportunity Assessment

Despite the constraints identified above, the site offers several extraordinary opportunities for playground development. The massive available open space (46,614 m²) provides an exceptionally generous footprint relative to the student population. The active play area requirement for 230 students is 80 m² per 50 students, scaled proportionally to **368 m² minimum** for the active zone, plus an additional **120 m² quiet/shade zone** (20 percent of total play area, accounting for the larger enrolment). The total minimum play infrastructure area is therefore approximately 488 m². A dedicated pre-primary play zone of approximately 550 m² has been designated within the larger compound, well within the available space. This enormous surplus allows for the most generous equipment spacing, widest safety zones, and clearest future expansion areas of any programme school.

The exceptional 60–70 percent tree canopy coverage is the single most significant asset on this site. Unlike other programme schools where tree canopy ranges from 20–40 percent, Assosa Primary and Middle School benefits from a dense, mature tree belt that provides extensive natural shade across much of the compound. This canopy will reduce surface temperatures under play equipment by an estimated 10–15 °C during peak afternoon heat, significantly exceeding the shade provision available at other sites. The trees also contribute to sensory development, air quality improvement, erosion control, and visual aesthetics. All preserved trees within and adjacent to the play zone shall have root protection zones of 1.0 m radius maintained during construction.

The site's elevation at 1,570 m ASL provides excellent natural drainage potential. With a minimum 1 percent slope engineered across the play zones, gravity-based drainage can direct all surface water toward collection points along the fence line. The extensive open space allows for longer drainage swale runs without the space constraints present at smaller schools. This approach is cost-effective, low-maintenance, and fully appropriate for the local construction capacity.

The existing functional equipment (2 slides, 1 balance beam, 1 functional swing, 2 ladders) provides a solid foundation that can be retained, refurbished with anti-rust treatment (zinc-chromate primer and enamel topcoat), and incorporated into the new layout. The slides require repainting but are structurally sound. One swing requires maintenance to replace worn chain links. The removal of the nonfunctional merry-go-round will free an additional 18 m² ($\pi \times 2.3^2 / 4$ + safety zone) for new equipment installation. These retained items reduce procurement costs and shorten the construction timeline.

As the largest school in the programme, Assosa Primary and Middle School serves as a flagship demonstration site for the IRC SUIDAC playground initiative. The comprehensive design described herein represents Option A: Maximum Play Value, providing the full range of equipment types at quantities scaled to 230 students — the highest in the programme.

2.1 Design Criteria and Scaling

All concept designs have been developed in accordance with the non-negotiable design standards specified in the IRC SUIDAC Playground Design Manual. Equipment dimensions, safety zone clearances, surface treatment depths, and material specifications follow the prescribed values exactly. The following scaling calculations apply to Assosa Primary and Middle School's enrolment of 230 students:

- **Active Play Zone:** $80 \text{ m}^2 \text{ per } 50 \text{ students} \times (230/50) = 368 \text{ m}^2$
- **Quiet/Shade Zone:** 20% of total play area $\approx 0.20 \times 550 = 110 \text{ m}^2$ (includes tree canopy integration)
- **Pathways:** Estimated $220 \text{ m} \times 1.2 \text{ m} = 264 \text{ m}^2$ (extended for larger zone)
- **Minimum Slides Required:** $1 \text{ per } 50 \text{ students} \times (230/50) = 5 \text{ slides}$ (2 existing + 3 new)
- **Minimum Swings Required:** $1 \text{ per } 30 \text{ students} \times (230/30) = 8 \text{ swing seats}$ (3 existing + 5 new)
- **Minimum See-Saws:** $1 \text{ per } 50 \text{ students} \times (230/50) = 5 \text{ see-saws}$ (all new)

Material Treatment Requirements: All metal components shall receive zinc-chromate primer followed by enamel topcoat (2 coats). All timber and bamboo shall undergo boron-based anti-termite treatment via 24-hour soak before installation. River sand surfacing shall be washed and graded to 150–300 mm depth under all equipment. Pathways shall use fine gravel at 100 mm compacted depth.

2.2 Option A — Maximum Play Value (Selected)

Given the largest student population of 230 children and the extraordinarily generous 46,614 m² of available open space, **Option A: Maximum Play Value** has been selected as the recommended and only appropriate design for Assosa Primary and Middle School. As the flagship school in the programme, this site receives the most comprehensive equipment package, with all quantities scaled to the 230-student enrolment. This option maximises the variety and quantity of play equipment within a safe, well-organised layout that provides a full range of developmental play experiences — gross motor, balance, climbing, social, and imaginative play — while maintaining full compliance with all safety standards.

2.2.1 Complete Equipment List

ID	Equipment	Dimensions (L×W×H)	Qty	Status	Safety Zone
E1	Merry-Go-Round	2.3m dia × 1.0m H	1	New (replaces removed)	2.0m radius
E2	See-Saw	4.0m L × 1.0m H	5	All new	8.0m × 3.0m total
E3	Slide	3.0m W × 2.0m H	5	2 existing + 3 new	2.0m runoff
E4	Chain Swing	2.5m × 2.5m × 2.5m H	5	2 existing + 3 new bays	2.0m all around
L1	Balance Beam	3.5m L × 0.25m H	3	1 existing + 2 new	1.0m sides
L2	Ladder	2.5m L × 0.6m W × 2.0m H	4	2 existing + 2 new	1.5m radius
L3	Tunnel Crawl	0.7m dia × 2.5m L	3	All new	1.0m sides
L4	Stepping Stones	5 stones, 0.3m dia × 0.15m H	3 sets	All new	0.5m sides

Table 2. Complete equipment schedule for Option A — Maximum Play Value.

2.2.2 Surface Area Calculations

Each equipment piece requires a defined surface area that includes the equipment footprint plus the mandatory safety zone. The following calculations use the exact dimensions from the equipment specifications, scaled to the 230-student enrolment:

Equipment	Footprint (m ²)	Safety Zone (m ²)	Total Surface (m ²)	Surface Type
E1 Merry-Go-Round (×1)	$4.2 (\pi \times 2.3^2 / 4)$	$12.6 (\pi \times 4.3^2 / 4 - \pi \times 2.3^2 / 4)$	16.8	River sand
E2 See-Saw (×5)	$5 \times 4.0 \times 0.5 = 10.0$	$5 \times (8.0 \times 3.0 - 4.0) = 100.0$	110.0	River sand
E3 Slide (×5)	$5 \times 3.0 \times 1.5 = 22.5$	$5 \times 6.0 = 30.0$	52.5	River sand
E4 Chain Swing (×5)	$5 \times 6.25 = 31.3$	$5 \times (6.25\pi / 4 \times (4.5/2.5)^2 - 6.25) \approx 57.0$	88.3	River sand
L1 Balance Beam (×3)	$3 \times 3.5 \times 0.25 = 2.6$	$3 \times 3.5 \times 2.0 = 21.0$	23.6	River sand
L2 Ladder (×4)	$4 \times 2.5 \times 0.6 = 6.0$	$4 \times (\pi \times 4.0^2 / 4 - 1.5) \approx 42.0$	48.0	River sand
L3 Tunnel Crawl (×3)	$3 \times 2.5 \times 0.7 = 5.3$	$3 \times 2.5 \times 2.0 = 15.0$	20.3	River sand
L4 Stepping Stones (×3 sets)	$3 \times 5 \times 0.3 \times 1.2 = 5.4$	$3 \times 6.0 = 18.0$	23.4	River sand

Table 3. Surface area calculations per equipment type (Option A).

Total equipment surface area: 382.9 m² of washed river sand at 150–300 mm depth. This exceeds the minimum active play zone requirement of 368 m² and, when combined with the interstitial circulation spaces between equipment zones, provides an effective active play area of approximately 490 m². The total dedicated pre-primary play zone is designed at approximately 550 m², accommodating the 382.9 m² equipment surface, circulation pathways of approximately 50 m², and the 110 m² quiet/shade zone integrated beneath the exceptional tree canopy.

2.2.3 Safety Zone Verification

All safety zones have been verified against the mandatory minimum clearances. The following checks confirm that no equipment safety zone overlaps with building walls, fence lines, or adjacent equipment zones:

Check Item	Standard	Design Value	Status
Fall zone around moving equipment	≥ 2.0 m	2.0 m (all E1–E4)	PASS
Clearance from fence line	≥ 1.5 m	1.5 m maintained	PASS
Clearance: swings/slides to buildings	≥ 3.0 m	3.5 m minimum	PASS
Pathway width	≥ 1.2 m	1.5 m main, 1.2 m secondary	PASS
Equipment height limit	≤ 2.5 m	Max 2.5 m (swing)	PASS
Safety zone overlap (inter-equipment)	None permitted	All verified clear	PASS
Tree root protection zone	≥ 1.0 m radius	1.0 m maintained (60–70% canopy)	PASS

Table 4. Safety zone compliance verification matrix.

2.2.4 Supervision Sight-Line Analysis

Teacher supervision is a critical safety requirement for pre-primary playgrounds. The design ensures that all play zones are visible from the pre-primary classroom building entrance, identified from the KML building footprints as one of the 22 structures within the compound. The layout organises equipment in a semi-circular arrangement within the dedicated 550 m² pre-primary zone, with the highest equipment (swings at 2.5 m, slides at 2.0 m) positioned closest to the classroom entrance. Lower-profile equipment (balance beams, stepping stones) is placed toward the outer edge of the zone.

The maximum sight-line distance from the pre-primary classroom entrance to any equipment piece is approximately 30 metres, well within the recommended 50 m maximum for pre-primary supervision. No equipment is positioned behind buildings, within dense tree clusters where visibility is compromised, or in any location where a child could be obscured from view. The three tunnel crawls (L3) are placed in the central play area with 360-degree visibility, ensuring that children inside the tunnel can be observed from the entry and exit points. The exceptional tree canopy does not impede sight-lines as the design avoids placing equipment directly under the densest canopy clusters.

2.2.5 Drainage Management

The drainage strategy employs a 1 percent minimum slope across all play zones, graded from the building cluster perimeter toward the fence line. Surface water flows via shallow swales (150 mm deep, 300 mm wide) along the pathway network toward collection points at the southeastern and southwestern corners of the fence line. The extensive 46,614 m² of open space allows for longer, more gradual swale runs compared to smaller programme schools.

Under each equipment installation, the 150–300 mm layer of washed river sand serves as both a fall-attenuation surface and a percolation medium. The sand layer is underlain by a geotextile fabric separator that prevents mixing with the lateritic subsoil while allowing water to drain through. The exceptional tree canopy coverage (60–70 percent) also contributes to drainage by intercepting rainfall and reducing runoff volume by an estimated 30–40 percent compared to open-ground sites. This dual-function approach eliminates the need for sub-surface drainage pipes and can be maintained by community labour through periodic sand loosening and top-up every 6–12 months.

2.3 Option B — Minimum Viable / Phased (Alternative)

Option B provides a conservative alternative with core equipment only, suitable for budget-constrained implementation scenarios. Given the 230-student enrolment — the largest in the programme — Option B would include the slides (2 existing, refurbished), chain swings (1 functional retained, 1 repaired, 1 new bay), three see-saws (new), one tunnel crawl (new), and one set of stepping stones (new). All other equipment categories are deferred to Phase 2 expansion. The total equipment surface area under Option B is approximately 180 m², concentrated in a compact layout adjacent to the pre-primary classroom.

Option B reserves two clearly demarcated expansion zones: Zone E1 (eastern open area, approximately 300 m²) for future installation of additional slides, see-saws, merry-go-round, ladders, and balance beams; and Zone E2 (southern strip, approximately 200 m²) for future tunnel crawls, stepping stones, and an outdoor classroom/shade structure integrated with the tree canopy. These zones are marked with low timber bollards and surfaced with compacted fine gravel to prevent weed growth and maintain usability as informal play space in the interim.

However, given the scale of this school and its role as the flagship site in the programme, Option A (Maximum Play Value) is strongly recommended as the primary design. Option B is presented here only as a contingency should budget constraints prevent full implementation. The total estimated cost of Option B is approximately 45 percent of Option A, with Phase 2 expansion costing an additional 60 percent of the Phase 1 budget.

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Slides	2 (existing, refurbished)	3 new	5
Chain Swings	2 existing + 1 repaired	3 new bays	8 seats
See-Saws	3 new	2 new	5
Merry-Go-Round	—	1 new	1
Balance Beams	—	3 total	3
Ladders	2 existing + 1 new	1 new	4
Tunnel Crawl	1 new	2 new	3
Stepping Stones	1 set new	2 sets	3 sets
Quiet/Shade Zone	60 m ² (under tree canopy)	50 m ² expansion	110 m ²

Table 5. Phased equipment plan for Option B — Minimum Viable.

3.1 Master Layout Plan Description

The master layout for Assosa Primary and Middle School organises the pre-primary playground into five distinct functional zones within a dedicated ~550 m² area, separated from the older primary and middle school students. The layout prioritises teacher sight-lines, natural drainage, age-appropriate zoning, and maximum integration with the exceptional 60–70 percent tree canopy that defines this site.

Zone 1 — Transition Area (Pre-Primary Classroom Perimeter): A 3.0 m wide paved transition strip surrounds the pre-primary classroom building(s), providing dry, safe access between classrooms and the play areas. This strip is surfaced with compacted fine gravel (100 mm depth) over geotextile fabric. No equipment is installed within this zone. The transition area connects directly to the 1.5 m main pathway system.

Zone 2 — Active Play Area (Central): The primary play zone occupies approximately 390 m² within the dedicated pre-primary area. Equipment is arranged in a semi-circular pattern, with high-activity equipment (5 swings at 2.5 m, 5 slides at 2.0 m, 1 merry-go-round) positioned toward the north (closer to the classroom entrance) and lower-profile equipment (3 balance beams, 3 sets stepping stones) toward the south. Five see-saws are placed on the western flank in a staggered row formation. A 1.5 m wide main pathway bisects the active zone from north to south, providing both circulation and emergency access. The generous spacing available at this large school allows all inter-equipment safety zones to be maintained at the maximum specified distances.

Zone 3 — Climbing and Exploration (Eastern Strip): Four ladders (L2) and three tunnel crawls (L3) are positioned along the eastern side of the active play area, partially integrated with the tree canopy for natural shade. A secondary 1.2 m pathway connects this zone to the main pathway. The ladders are metal frame structures at 2.0 m height, and the tunnel crawls are metal pipe sections at 700 mm diameter, both suitable for the 5–6 year age group.

Zone 4 — Quiet/Shade Zone (Tree Canopy Integration): The exceptional 60–70 percent tree canopy is leveraged to create a 110 m² quiet zone directly beneath the densest canopy coverage, adjacent to the play area. Simple log seating and a small shade platform are installed beneath the tree canopy for reading, rest, and low-activity play. This zone is separated from the active play area by a 1.2 m gravel pathway that serves as a physical and visual buffer. The dense canopy overhead eliminates the need for artificial shade structures in this zone.

Zone 5 — Drainage and Service (Perimeter): Linear drainage swales run along the perimeter of the play zone, collecting water from all zones via the graded slope. A 1.5 m wide service path between the drainage swales and the outermost equipment allows maintenance access without entering the play zones. The service path connects to the main entrance road for material delivery.

3.2 Equipment Placement Schedule

The following table provides relative placement coordinates for each equipment installation. Coordinates are referenced from the southwestern corner of the pre-primary play zone (origin: 0, 0), with X increasing eastward and Y increasing northward. All positions maintain the required safety clearances from buildings, fences, tree root zones, and adjacent equipment.

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E1	Merry-Go-Round	1	2	(24, 48)	N/A (circular)	Replaces removed unit
E2a	See-Saw	1	2	(6, 38)	E-W axis	New, 3.0 m from fence
E2b	See-Saw	1	2	(6, 30)	E-W axis	New, 6.0 m from E2a
E2c	See-Saw	1	2	(6, 22)	E-W axis	New, 6.0 m from E2b
E2d	See-Saw	1	2	(6, 14)	E-W axis	New, 6.0 m from E2c
E2e	See-Saw	1	2	(6, 6)	E-W axis	New, 6.0 m from E2d
E3a	Slide (existing)	1	2	(22, 42)	S-facing	Refurbished, repainted
E3b	Slide (existing)	1	2	(30, 42)	S-facing	Refurbished, repainted
E3c	Slide (new)	1	2	(38, 42)	S-facing	5.0 m from E3b
E3d	Slide (new)	1	2	(46, 42)	S-facing	5.0 m from E3c
E3e	Slide (new)	1	2	(54, 42)	S-facing	5.0 m from E3d
E4a	Chain Swing	1	2	(18, 46)	N/A	Existing, functional
E4b	Chain Swing	1	2	(32, 48)	N/A	Existing, maintenance req.
E4c	Chain Swing	1	2	(42, 48)	N/A	New bay
E4d	Chain Swing	1	2	(50, 48)	N/A	New bay
E4e	Chain Swing	1	2	(58, 48)	N/A	New bay
L1a	Balance Beam	1	2	(20, 20)	N-S axis	Existing
L1b	Balance Beam	1	2	(28, 20)	N-S axis	New
L1c	Balance Beam	1	2	(36, 20)	N-S axis	New
L2a	Ladder	1	3	(54, 36)	N-S axis	Existing
L2b	Ladder	1	3	(60, 36)	N-S axis	Existing
L2c	Ladder	1	3	(54, 26)	N-S axis	New
L2d	Ladder	1	3	(60, 26)	N-S axis	New
L3a	Tunnel Crawl	1	3	(52, 16)	E-W axis	New, metal pipe
L3b	Tunnel Crawl	1	3	(58, 16)	E-W axis	New, metal pipe
L3c	Tunnel Crawl	1	3	(64, 16)	E-W axis	New, metal pipe
L4a	Stepping Stones	1 set	2	(18, 8)	N-S line	5 stones, 400mm spacing
L4b	Stepping Stones	1 set	2	(32, 8)	N-S line	5 stones, 400mm spacing
L4c	Stepping Stones	1 set	2	(46, 8)	N-S line	5 stones, 400mm spacing

Table 6. Equipment placement schedule with relative coordinates (Option A).

3.3 Grading and Earthwork Notes

The existing site gradient slopes gently from the building cluster toward the fence perimeter, consistent with the natural topography of Assosa at 1,570 m ASL. The 22-building campus and extensive open space suggest a generally flat terrain with minor undulations. Minor grading will be required to achieve the specified 1.0 percent drainage slope across the 550 m² pre-primary play zone.

Earthwork Item	Description	Estimated Volume	Method
Slope correction	Grade play zone to 1.0% slope from north (classroom) to south (fence line)	≈ 18 m ³ cut/fill	Hand-compacted laterite
Equipment platforms	Level pads under each of 29 equipment installations, 100 mm above surrounding grade	≈ 12 m ³ fill	Compacted gravel base
Drainage swales	150 mm deep × 300 mm wide swales along perimeter of play zone, ~80 m total length	≈ 4 m ³ excavation	Hand-dug, geotextile lined
Pathway sub-base	Fine gravel, 100 mm compacted over geotextile, 220 m total length (1.5 m main + 1.2 m secondary)	≈ 33 m ³	Imported gravel, hand-compacted
Sand surfacing	Washed river sand, 200 mm avg depth under all equipment zones (383 m ²)	≈ 77 m ³	Imported, levelled by hand
Tree protection mounds	Circular earth mounds around preserved tree trunks within play zone (est. 15 trees, 1 m radius)	≈ 4 m ³	Hand-shaped

Table 7. Grading and earthwork schedule.

Total estimated earthwork volume: 148 m³ (cut/fill). All earthwork shall be executed by community labour using hand tools (shovels, rakes, hand tampers). No mechanical equipment is required. The cut-and-fill balance ensures that excavated material from the drainage swales is reused as fill for slope correction and equipment platforms, minimising material transport within the large 50,662 m² compound.

A compaction standard of 95 percent Modified Proctor Density shall be achieved for all filled areas using hand tampers. Quality assurance shall be performed by the site supervisor using a simple probe test (steel rod penetration resistance) at 5 m intervals across all graded areas. Given the larger play zone area, compaction verification will require approximately twice the number of probe tests compared to smaller programme schools.

3.4 Planting Plan

The planting strategy for Assosa Primary and Middle School focuses on four objectives: supplementary shade provision within equipment zones where existing canopy is thin, erosion control on graded slopes, aesthetic enhancement, and demarcation of the pre-primary play zone from the broader campus. The exceptional 60–70 percent existing canopy reduces the need for new shade trees, allowing the planting budget to focus on lower-profile species for boundary definition and erosion control.

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
Ficus sycomorus (Shola)	Deciduous shade tree	3	Along northern edge of play zone where canopy is thinner	Supplementary shade for swing zone	10 m

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
<i>Terminalia brownii</i> (Dhithessa)	Semi-deciduous tree	2	Adjacent to drainage swales at southeast corner	Root stabilisation for swale banks	10 m
<i>Codiaeum variegatum</i> (local ornamental)	Low shrub	20	Along pathway edges and play zone boundary, alternating sides	Visual definition, zone demarcation	1.5 m
<i>Cynodon dactylon</i> (Bermuda grass)	Ground cover	120 m ²	Quiet zone under tree canopy and expansion areas	Erosion control, dust suppression	Broadcast seed
<i>Euphorbia tirucalli</i> (local hedge)	Low hedge (0.4 m)	50 m length	Boundary between pre-primary zone and older students' area	Visual separation, safety demarcation	0.5 m (linear)
<i>Bougainvillea spectabilis</i> (local ornamental)	Flowering shrub	8	Entrance to pre-primary zone and quiet zone perimeter	Colour, sensory stimulation	3 m

Table 8. Planting schedule — locally sourced species.

The existing mature tree canopy (60–70 percent) shall be preserved in its entirety. No pruning shall be performed without arborist assessment. Root protection zones of 1.0 m radius around each trunk shall be maintained during grading and construction. Circular earth mounds shall be built around the trunks within the play zone to prevent soil compaction and root damage from foot traffic. An estimated 15 trees fall within or adjacent to the 550 m² play zone and require individual protection measures.

3.5 Maintenance Framework

Long-term playground sustainability depends on a clear maintenance protocol that can be executed by school staff and community volunteers with basic tools. Given the larger scale of this playground (29 equipment installations across 550 m²), the maintenance burden is correspondingly greater than at smaller programme schools. The following framework defines maintenance tasks, frequencies, responsible parties, and material requirements.

Task	Frequency	Responsible	Materials/Tools	Notes
Sand loosening and top-up	Monthly	School groundskeeper	Rake, wheelbarrow, river sand	Check depth across 383 m ² , replace if below 150 mm
Metal equipment rust check	Quarterly	School maintenance lead	Wire brush, primer, enamel paint	Inspect all 5 slides, 5 swings, 4 ladders
Timber/bamboo termite check	Bi-annually (pre/post rain)	Community volunteer team	Boron solution, brush	Re-soak if soft spots detected
Fastener and joint inspection	Monthly	School groundskeeper	Wrench set, replacement bolts	Check all swing chains, slide anchors, see-saw pivots
Drainage swale clearing	After each rain event	School staff	Shovel, hand trowel	Clear debris from ~80 m of swale network

Task	Frequency	Responsible	Materials/Tools	Notes
Pathway gravel replenishment	Bi-annually	Community volunteer team	Fine gravel, rake	Top up 220 m of pathways to 100 mm, re-compact
Tree watering (new plants)	Daily for first 3 months	Student water monitors (supervised)	Watering cans	Morning watering, 5 L per plant per day
Fence integrity check	Quarterly	School maintenance lead	Pliers, wire, replacement posts	Check 44-vertex fence for gaps, rust, loose wires
Play zone boundary check	Monthly	School groundskeeper	Paint, brush	Repaint Euphorbia hedge boundary markers as needed

Table 9. Maintenance schedule and responsibility matrix.

3.6 Material Sourcing — Local Availability

All materials specified in this design are available within a 50 km radius of Assosa city. This local sourcing requirement supports the community-labour construction model and minimises procurement lead times and transportation costs. The larger scale of this school means proportionally greater material quantities compared to smaller programme schools.

Material	Source Location	Distance from Site	Estimated Cost Rate
Washed river sand	Blue Nile riverbed, south of Assosa	25 km	ETB 800/m ³
Fine gravel (pathways)	Assosa municipal quarry	8 km	ETB 650/m ³
Bamboo (construction grade)	Mao-Komo district forests	40 km	ETB 150/pole
Timber (Eucalyptus posts)	Assosa plantation nursery	12 km	ETB 200/post
Metal pipe (600–800 mm)	Assosa metalworks workshop	5 km	ETB 3,500/piece
Zinc-chromate primer	Assosa hardware shops	3 km	ETB 450/gallon
Enamel topcoat paint	Assosa hardware shops	3 km	ETB 550/gallon
Boron treatment chemical	IRC supply chain (Addis Ababa)	Air freight	Included in IRC logistics
Geotextile fabric	IRC supply chain	Air freight + 5 km	Included in IRC logistics
Chain and hardware (swings)	Assosa metalworks / Bahir Dar	5–180 km	ETB 2,000/set

Table 10. Local material sourcing schedule.

End of Design Document — Assosa Primary and Middle School (School 4 of 6)

This document was prepared under the IRC SUIDAC programme for Cities Alliance donor review. All dimensions, calculations, and specifications are site-specific to Assosa Primary and Middle School and should not be applied to other school sites without independent verification.